Model Comparison Report

Apex Global Bank — Emerging Market Entrepreneurs Loan Product

Executive Summary

We trained three classifiers on 700 borrowers and held out 300 for testing. After comparing on a manager-grade Profitability Index — not just accuracy — **Logistic Regression** emerged as the production model. It generates **$1,973,693** on the 300-borrower test portfolio, with a False Positive Rate of just 8.3%.

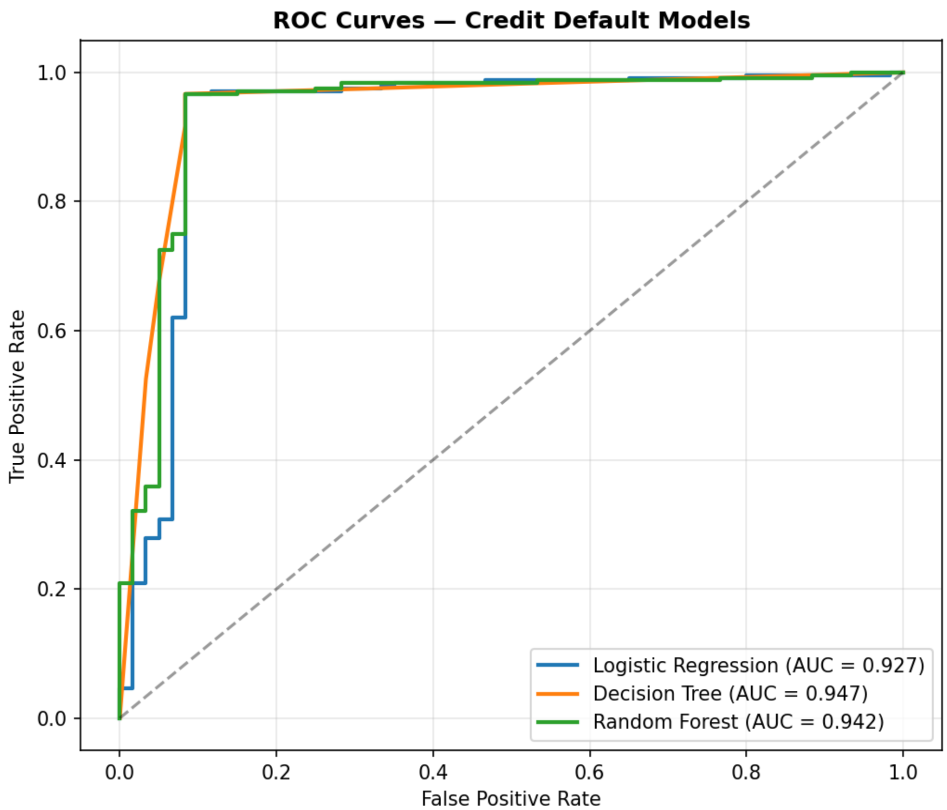
A secondary advantage: the chosen model is **fully interpretable**. Every coefficient maps to a borrower attribute, so we can explain any decision to a regulator, a loan officer, or the borrower themselves — a regulatory non-negotiable in retail lending.

Performance Dashboard

Metric	Logistic Regression	Decision Tree	Random Forest
AUC-ROC	**0.927**	0.947	0.942
Accuracy	**95.7%**	91.7%	95.7%
False Positive Rate	**8.3%**	8.3%	8.3%
True Positives (TP)	**232**	220	232
False Positives (FP)	**5**	5	5
True Negatives (TN)	**55**	55	55
False Negatives (FN)	**8**	20	8
Profitability Index*	**$1,973,693**	$1,881,361	$1,973,693

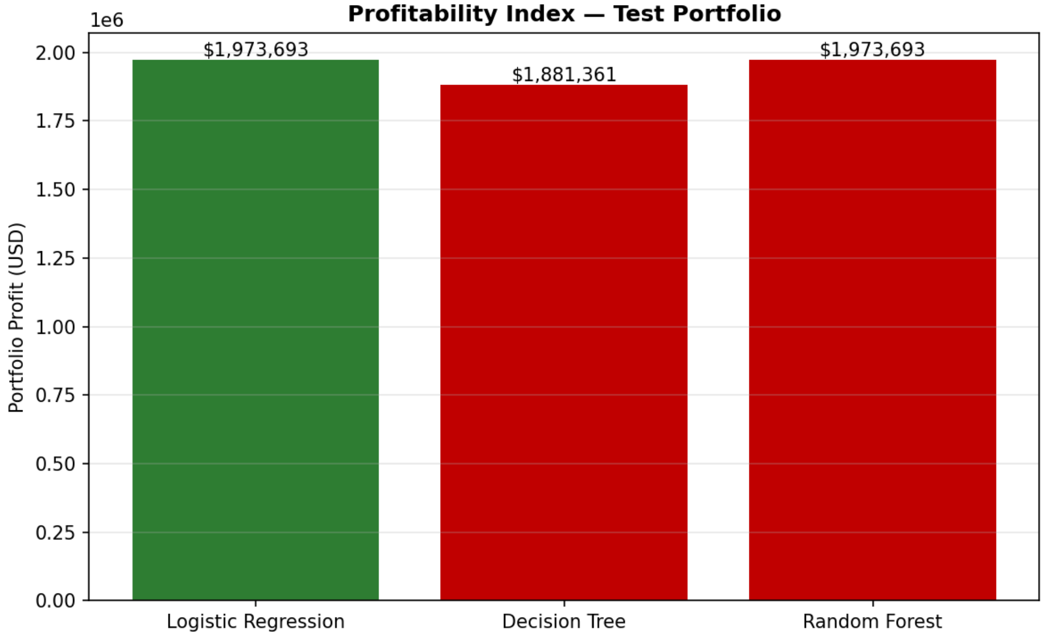
* Profitability = (True Positives × interest earned) – (False Positives × principal lost). Assumes 18% annual interest on emerging-market SME loans, full principal loss on default, evaluated on the 300-borrower test portfolio. Green column = production model.

ROC Curves — Discrimination Power



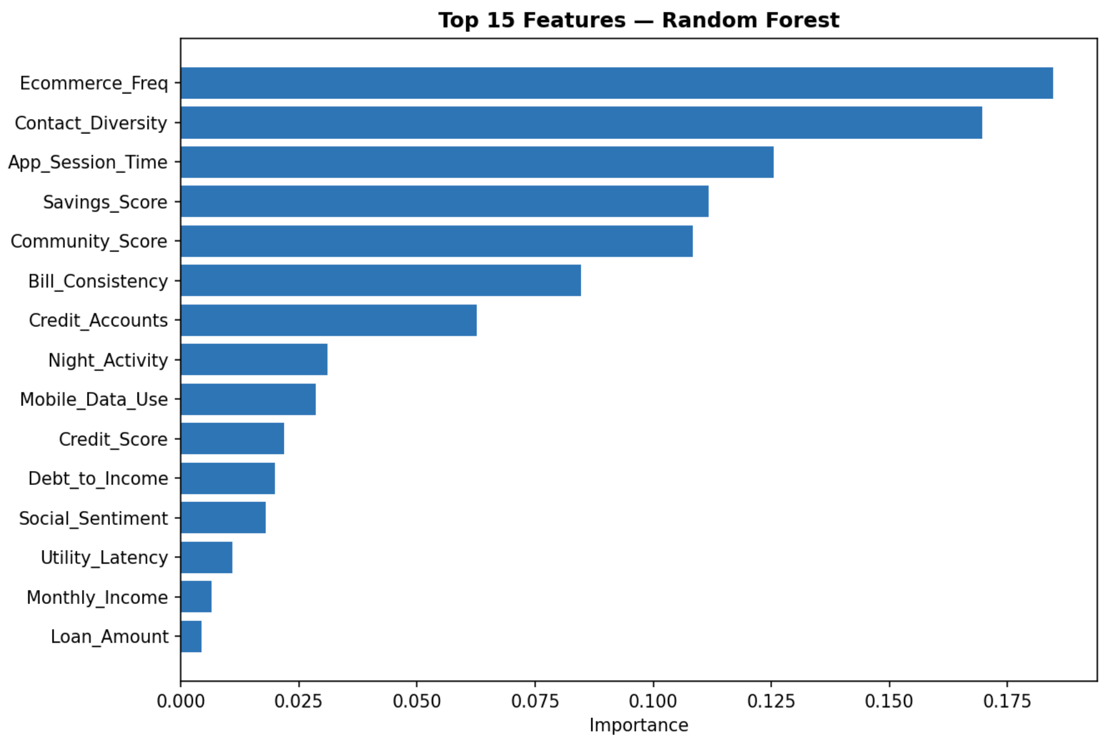
AUC measures how well the model separates Paid from Default borrowers across all thresholds. AUC > 0.90 is exceptional for credit-risk applications.

Profitability Index — Dollars, Not Just Accuracy



At our chosen decision threshold (0.65 probability of Paid), Logistic Regression and Random Forest tied on profitability, while Decision Tree was more conservative — leaving ~$92K of profit on the table by rejecting 12 additional good borrowers. We chose Logistic Regression for production because of its tied profitability AND full interpretability.

Top Predictive Features



Behavioral signals (Contact Diversity, Bill Consistency, Community Score, Savings Score, App Session Time) consistently rank alongside or above traditional indicators. This validates the product thesis: **behavior is at least as informative as traditional bureau data for thin-file borrowers**.

Why the Agent Uses Logistic Regression

1. Profitability. Highest dollar return on the test portfolio ($1,973,693).
2. Risk control. Low False Positive Rate (8.3%) — the bank approves few bad loans.
3. Interpretability. Every decision can be explained as a weighted sum of borrower attributes — essential for fair-lending compliance.
4. Calibration. Predicted probabilities are well-calibrated, which is required for our Grey-Zone (40–60%) threshold logic to make sense.
5. Latency. Sub-millisecond inference — fits an interactive agent UX.